Ragnork RAG System Technical Report
Author: Architecture Team

1. Introduction
The Ragnork system implements a NotebookLM-style Retrieval-Augmented Generation application.
Users upload PDFs, and all answers are strictly grounded in source documents.

2. Database Architecture
Postgres is the system of record for all metadata, elements, and text chunks.
Qdrant is used for dense vector indexing. Tenant isolation is enforced at the database layer using notebook_id a

3. Performance Benchmark Table

Table 1: System Latency Metrics
Stage | Target Latency | P99 Latency
Ingestion Parse | 500ms | 1200ms
Vector Retrieval | 50ms | 120ms
LLM Generation | 2000ms | 3500ms

4. Citation Standards
Every generated answer carries explicit citation handles [1], [2] referencing exact page numbers and bounding b